

4.3 HVAC System Completion

Installation, startup and commissioning of the complete HVAC system — the measurements that prove it runs efficiently, safely and as designed.

Project Address: _____

Project Name: _____ Permit Number: _____

HVAC Contractor: _____ License Number: _____

Contact Phone: _____

PURPOSE: This document verifies proper installation, startup and commissioning of the complete HVAC system. Proper commissioning ensures the system operates efficiently, safely and as designed.

EQUIPMENT INSTALLATION

Indoor Unit (Furnace / Air Handler)

Make/Model: _____ Serial Number: _____

BTU Input/Output: _____ AFUE Rating: _____

Location: _____ Installation Date: _____

FUEL TYPE: ☐ Natural Gas ☐ Propane ☐ Electric ☐ Oil

- ☐ Unit installed level and properly supported
- ☐ Proper clearances maintained (front, sides, top per manufacturer)
- ☐ Unit properly secured to platform or hung from joists
- ☐ Gas line connected properly (if gas) — no leaks
- ☐ Gas shut-off valve within 6 feet (if gas)
- ☐ Gas line pressure tested (if gas)
- ☐ Electrical connection made — proper voltage and amperage
- ☐ Electrical disconnect within sight of unit
- ☐ Condensate drain line installed with proper slope

- ☐ Condensate drain terminates appropriately
- ☐ Condensate trap installed (if required)
- ☐ Secondary drain pan installed (if in attic/living space)
- ☐ Secondary drain pan alarm/sensor installed
- ☐ Combustion air intake adequate (if gas)
- ☐ Vent pipe properly installed (if gas)
- ☐ Vent pipe properly supported and sloped
- ☐ Vent termination proper distance from windows/openings
- ☐ Filter access accessible
- ☐ Blower door/access panels properly secured

Outdoor Unit (Condensing Unit / Heat Pump)

Make/Model: _____ Serial Number: _____

Tonnage: _____ SEER Rating: _____

Refrigerant Type: _____ Location: _____

TYPE: ☐ Air Conditioner ☐ Heat Pump

- ☐ Unit installed on level pad (concrete or composite)
- ☐ Pad stable and properly sized
- ☐ Unit level side-to-side and front-to-back (within 1/4")
- ☐ Proper clearances maintained (12" sides, 24" service side, 60" top)
- ☐ No vegetation or obstructions blocking airflow
- ☐ Refrigerant lines connected and insulated
- ☐ Line set insulation complete — no exposed copper
- ☐ Line set properly secured to structure
- ☐ Electrical disconnect installed within sight of unit
- ☐ Electrical disconnect proper rating (check data plate)
- ☐ Whip (flexible conduit) to unit installed properly

- ☐ Electrical connections tight at contactor and terminals
- ☐ Condenser fan rotates freely by hand (power off)
- ☐ Coil fins undamaged and clean
- ☐ Service valves accessible

DUCTWORK COMPLETION

- ☐ All supply registers installed
- ☐ All supply registers properly secured
- ☐ Register boot connections sealed with mastic
- ☐ All return grilles installed
- ☐ Return grille connections sealed
- ☐ Main trunk line connections sealed
- ☐ All branch take-offs sealed
- ☐ Flexible duct properly supported (every 4-5 feet)
- ☐ Flexible duct not kinked or crushed
- ☐ Ductwork insulation complete (if required)
- ☐ No disconnected ductwork
- ☐ Filter installed — correct size recorded below
- ☐ Filter access accessible
- ☐ Return air pathways adequate (no blocked returns)
- ☐ Access panels installed for dampers and controls

Filter size installed: _____

Register / Grille Schedule

Room	Supply Register Size	Qty	Return Grille Size	Qty	Notes

Register / Grille Schedule

Room	Supply Register Size	Qty	Return Grille Size	Qty	Notes

THERMOSTAT INSTALLATION

Make/Model: _____ Location: _____

Number of Zones: _____

TYPE: ☐ Programmable ☐ Non-Programmable ☐ Smart/WiFi

- ☐ Thermostat located on interior wall (not exterior)
- ☐ Location away from direct sunlight
- ☐ Location away from heat sources (lamps, appliances)
- ☐ Location away from drafts (doors, windows)
- ☐ Thermostat mounted level at 52-60" height
- ☐ All wires properly labeled
- ☐ All wire connections tight
- ☐ Thermostat wired correctly — tested with system
- ☐ Programmable thermostat programmed (if applicable)
- ☐ Smart thermostat connected to WiFi (if applicable)
- ☐ Homeowner trained on thermostat operation
- ☐ Manual/instructions left with homeowner

SYSTEM COMMISSIONING

Professional commissioning must be performed by a qualified HVAC technician:

Commissioning Technician: _____ Company: _____

License Number: _____ Commissioning Date: _____

Refrigerant Charging (Air Conditioner / Heat Pump)

Measurement	Reading / Value
Refrigerant type	
Charge weight (lbs)	
Charging method	☐ Subcooling ☐ Superheat ☐ Weigh-In

Refrigerant Charging (Air Conditioner / Heat Pump)

Measurement	Reading / Value
Outdoor temperature (°F)	
Indoor temperature (°F)	
Indoor relative humidity (%)	

Subcooling Method (preferred for TXV systems)

Measurement	Reading / Value
High side pressure (PSI)	
Saturation temperature (°F)	
Liquid line temperature (°F)	
Subcooling (°F)	
Target subcooling per manufacturer	
Subcooling within spec	☐ Yes ☐ No

Superheat Method (fixed orifice / piston systems)

Measurement	Reading / Value
Low side pressure (PSI)	
Saturation temperature (°F)	
Suction line temperature (°F)	
Superheat (°F)	
Target superheat per manufacturer	
Superheat within spec	☐ Yes ☐ No

Airflow Verification

Measurement	Reading / Value
Blower speed setting	
Supply air temperature (°F)	
Return air temperature (°F)	
Temperature split (°F)	
Target split (cooling: 15–20°F)	
Airflow CFM (if measured)	
Target airflow (400 CFM/ton)	
Airflow adequate	☐ Yes ☐ No

Gas Furnace Checks (if applicable)

Measurement	Reading / Value
Gas type	☐ Natural Gas ☐ Propane
Manifold pressure (in. w.c.)	
Target manifold pressure	
Temperature rise (°F)	
Target temperature rise range	
Flame appearance	☐ Blue/Stable ☐ Other — describe in notes
Carbon monoxide test (PPM)	
CO level acceptable (<10 PPM)	☐ Yes ☐ No
Vent draft adequate	☐ Yes ☐ No ☐ N/A

OPERATIONAL TESTING

Heating Mode Test

- ☐ Thermostat set to HEAT mode
- ☐ Temperature set above room temperature
- ☐ Furnace/heat pump starts within 1-2 minutes
- ☐ Ignition sequence normal (if gas)
- ☐ Burner flames stable and blue (if gas)
- ☐ Blower starts after heat-up delay
- ☐ Warm air from all supply registers
- ☐ Airflow adequate at all registers
- ☐ System runs continuously without cycling
- ☐ No unusual noises during operation
- ☐ System reaches temperature setpoint and cycles off
- ☐ Blower continues for cool-down period
- ☐ No error codes or warning lights

Cooling Mode Test

- ☐ Thermostat set to COOL mode
- ☐ Temperature set below room temperature
- ☐ Outdoor unit starts (compressor and fan)
- ☐ Indoor blower starts
- ☐ Cool air from all supply registers
- ☐ Temperature drop 15–20°F (supply vs. return)
- ☐ Condensate draining properly
- ☐ No water leaks from indoor unit
- ☐ Outdoor unit running smoothly — no unusual noises
- ☐ System runs continuously without short-cycling
- ☐ System reaches temperature setpoint and cycles off
- ☐ No error codes or warning lights

General System Operation

- ☐ Thermostat controls system properly
- ☐ Fan mode AUTO and ON both work
- ☐ System cycles properly — not short-cycling
- ☐ No vibration in ductwork during operation
- ☐ No whistling or whooshing sounds from registers
- ☐ All rooms receiving adequate airflow
- ☐ Return air adequate — no negative pressure issues
- ☐ Emergency heat operates (if heat pump) — or N/A
- ☐ Reversing valve operates (if heat pump) — or N/A
- ☐ Outdoor unit clean and free of debris

AIRFLOW BALANCING

Verify adequate airflow to all rooms:

Room-by-Room Balance Record

Room	Supply Register	Airflow	Temperature	Damper Adjustment	Notes
		☐ Good ☐ Low			
		☐ Good ☐ Low			
		☐ Good ☐ Low			
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		☐ Good ☐ Low			

☐

All rooms receiving airflow

- ☐ Hot/cold spots identified and addressed
- ☐ Dampers adjusted for balanced airflow
- ☐ Temperature variation between rooms minimal (within 3–4°F)

FINAL HVAC INSPECTION

Inspection Requested Date: _____ Inspection Scheduled Date: _____

Inspector Name: _____

Inspection Date: _____ Inspection Time: _____

Pre-Inspection Checklist

- ☐ All equipment properly installed
- ☐ All ductwork complete and sealed
- ☐ System commissioned by qualified technician
- ☐ Refrigerant charge verified
- ☐ Heating and cooling both tested
- ☐ Thermostat functioning properly
- ☐ All required disconnects installed
- ☐ Proper clearances maintained
- ☐ Condensate drains functioning
- ☐ Work completed per approved plans

INSPECTION RESULT: ☐ PASSED — Certificate Issued ☐ FAILED — Corrections Required

Required Corrections

Corrections Completed Date: _____ Re-Inspection Date: _____

RE-INSPECTION RESULT: ☐ PASSED ☐ FAILED

WARRANTY & DOCUMENTATION

Equipment Warranty Period: _____ Parts Warranty Period: _____

Labor Warranty Period: _____ Compressor Warranty Period: _____

REGISTRATION REQUIRED: ☐ Yes ☐ No

REGISTRATION COMPLETED: ☐ Yes ☐ No ☐ N/A

Registration Date: _____

Documentation Provided to Homeowner

- ☐ Equipment owner's manuals (indoor and outdoor units)
- ☐ Thermostat manual and programming guide
- ☐ Warranty information and registration
- ☐ Filter size and replacement schedule
- ☐ Maintenance schedule and recommendations
- ☐ Contractor contact information for service
- ☐ As-built duct layout (if available)
- ☐ Commissioning report with all measurements

Sign-Off

HVAC Technician Signature: _____ Date: _____

Owner/Builder Signature: _____ Date: _____